

Odysseus CUDA Setup Guide

Faa Odysseus til at kore med NVIDIA GPU
og llama.cpp CUDA-support

GTX 1080 Ti | CUDA 12.4 | Docker | Odysseus

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1. Forudsætninger

For at komme i gang skal du have følgende på din maskine:

- NVIDIA GPU med driver installeret
- Docker + docker compose
- NVIDIA Container Toolkit (nvidia-container-toolkit)

Tjek at alt er klar:

```
nvidia-smi
docker info | grep -i nvidia
nvidia-ctk --version
```

2. Klon Odysseus

Hent Odysseus fra GitHub:

```
git clone https://github.com/.../odysseus.git
cd odysseus
```

3. Konfigurer .env

Kopier .env.example til .env og sæt:

```
COMPOSE_FILE=docker-compose.yml:docker/gpu.nvidia.yml
APP_BIND=0.0.0.0
APP_PORT=7200
```

4. Opdater Dockerfile

Skift base image:

```
FROM nvidia/cuda:12.4.1-devel-ubuntu22.04
```

CUDA-image har ikke Python — tilføj i apt-get:

```
python3 \
python3-pip \
python3-venv \
```

Ret pip install til pip3 install.

5. Opdater docker/entrypoint.sh

Erstat CUDA_HOME-blokken (linje 64-72 i original) med:

```
if [ -x /usr/local/cuda/bin/nvcc ]; then
    export CUDA_HOME=/usr/local/cuda
else
    for cu in \
        /app/.local/lib/python*/site-packages/nvidia/cu13 \
        /app/.local/lib/python*/site-packages/nvidia/cu12 \
        /app/.local/lib/python*/site-packages/nvidia/cuda_nvcc; do
        if [ -x "$cu/bin/nvcc" ]; then
            export CUDA_HOME="$cu"
            break
        fi
    done
fi
```

Dette sikrer at systemets CUDA (med nvcc + libcudart) prioriteres over pip-stubben.

6. Byg og start

```
docker compose build --no-cache odysseus
docker compose up -d
```

7. Verificer CUDA

```
docker compose exec odysseus nvcc --version
docker compose exec odysseus nvidia-smi
```

8. Installer llama-cpp-python

Gå til Odysseus → Cookbook → Dependencies → Installer llama-cpp-python[server].

Forventet output:

```
[odysseus] CUDA nvcc + cudart found – building
llama-server with CUDA (GPU) support...
```

9. Serve en model

Brug -ngl 99 for fuld GPU offload:

```
llama-server -m /app/models/mistral-7b.gguf \  
  --host 0.0.0.0 --port 8080 -ngl 99
```

10. Fejlfinding

Problem	Løsning
nvcc not found	Dockerfile mangler CUDA-base
CUDA runtime missing	Entrypoint overskriver CUDA_HOME
CUDA_HOME=/app/.local/	Entrypoint prioriterer pip-stub
Python not found	CUDA-image mangler Python
pip: not found	Brug pip3 i stedet